

BRAC University, Dhaka
Department of Computer Science and Engineering

CSE 446: Blockchain and Cryptocurrencies, Spring 2026, Quiz - 1

Marks: 20

Time: 20 Min

Name:	ID:	Section:
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1. Determine if the following sentences are true or false. For any false sentence, write its correct form. 4*1 = 4
 - i. Bloom filter always generates true positive results. (F)
Sometimes it can generate false positive results.
 - ii. A hash function accepts only inputs of fixed length. (F)
It accepts inputs of any variable length of input.
 - iii. In an asynchronous network, a message will be delivered to all nodes. (T)
 - iv. A software bug can be the source of a byzantine fault. (T)

2. Fill in the blanks 4*2 = 8
 - i. A candidate in RAFT becomes a follower when _ discovers current leader or a new term _
 - ii. A hash pointer contains _ location of the data _ & _ the hash of the data _
 - iii. A digital signature uses _ private _ key of the _ sender _
 - iv. Two properties of a consensus algorithm are _ any two of safety/consistency, liveness/availability and Fault tolerance _ & _

3. Briefly answer the following questions. 4*2 = 8
 - i. How can you ensure consensus in a replicated state machine?
By utilising a protocol that guarantees atomic broadcast.
 - ii. What happens when a vote is split in Raft?
The term ends without a leader.
 - iii. Why was Bitcoin introduced?
In order to dissolve the trust in a centralised system and replace with a distributed/decentralised trust.
 - iv. How to tackle a pre-image resistance attack for a value from a low min-entropy distribution?
By concatenating with a value taken from a high min-entropy distribution and then generating the hash.